🏥

Cura Health

Complete Technical & Clinical Documentation

VERSION	DATE	DISEASES IN DB
Launch-Locked v1.0	April 2026	16,000+ ICD-11 Coded

CLINICAL DOMAINS	INPUT CATEGORIES
23 Domains · 19 Tiles	19 Health Categories

⚠️ This document is for internal and stakeholder use only. The risk profile is a statistical screening tool, not a clinical diagnosis. Always consult a qualified physician.

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1 What Is the Risk Profile?

Purpose, scope, and what it delivers to the patient

The Cura Health Risk Profile is a **personalized, multi-factor health risk assessment** that tells a patient which body systems they should monitor based on their complete health profile. It is not a diagnosis. It is a signal — a smart, data-driven way to direct a patient's attention toward areas of elevated risk before they become problems.

Every time a patient fills in their personal details (age, gender, diet, medications, family history, vaccinations, and 15+ other categories), the system runs a full scoring computation against a database of **over 16,000 ICD-11 coded diseases**. The result is a personalized map of 17–19 body system tiles, each showing either **Worth monitoring** or **Looking good** .

💡 **Think of it like a smart health radar**

Just as a radar doesn't tell you exactly where a storm will hit — but tells you which direction to watch — the risk profile doesn't diagnose disease, but tells a patient which body systems deserve attention given everything we know about them.

2 Core Design Philosophy

The principles that drive every decision in this engine

🔴 **The Golden Rule**

A **false negative is dangerous**. A false positive is acceptable. A false negative means we tell a patient their heart health looks fine when it doesn't — that is a life-threatening miss. A false positive means we flag lungs for a healthy patient — at worst, they ask their doctor about it. When in doubt, we surface risk. We never suppress it.

Three Problems This Engine Solves

P1

Large domains always win

Cardiovascular has ~580 diseases. Injury/Toxicology has ~1,982. Purely summing scores means larger domains always dominate — not because the patient has more risk there, but because more diseases exist to collect the score.

P2

Family history contaminates scores

The DB's generic `famhist_*` columns boost ALL diseases equally. A patient with a father who has bone disease gets a cardiovascular boost instead of an MSK boost — completely wrong.

P3

Unmapped domains become silent misses

If a DB domain has no frontend tile, any patient scoring high there sees no warning — a direct false negative. With 909 diseases in "External Causes / Prevention", this is a real risk.

The Solutions

- **Problem 1 → Excess Scoring:** Subtract each domain's own age+gender baseline. Now every domain competes on how much this patient's specific factors push risk *above* their demographic baseline.

- **Problem 2 → Condition-Specific Family Routing:** Strip `famhist_*` from domain scoring entirely. Use a direct condition→domain map to boost exactly the right body system.
- **Problem 3 → 19 tiles covering all 23 domains:** Every single DB clinical domain maps to a visible frontend tile. Nothing falls through the cracks.

3 The Disease Database

The foundation: 16,000+ ICD-11 diseases, each encoded as a risk-factor vector

Table: `disease_risk_scores`

Each **row** is one disease. Each **column** is one risk factor. The cell value is a decimal (0.0 to ~0.8) representing how strongly that factor is associated with that disease.

```
-- Simplified structure of disease_risk_scores
disease_risk_scores
├── code                          -- ICD-11 code (e.g. "BA00")
├── title_text                    -- Disease name (e.g. "Type 2 Diabetes Mellitus")
├── clinical_domain               -- Body system (e.g. "Endocrine/Metabolic")
├── chapter                       -- ICD-11 chapter
├── age_18_to_25_years_risk       -- Risk weight for 18-25 year olds
├── male                          -- Risk weight for male gender
├── female                        -- Risk weight for female gender
├── comorb_diabetes_mellitus_rf   -- Boost if patient has diabetes
├── famhist_father_rf             -- Generic boost if father has any condition
├── diet_red_meat_rf              -- Boost if patient eats red meat frequently
... 200+ columns total, one per risk factor
```

Scale of the Database

Clinical Domain	Disease Count	Frontend Tile
Injury/Toxicology	1,982	Injury & Toxins
Congenital/Genetics	1,323	Genetics & Inherited
Oncology	1,247	Cell Health
Undifferentiated Clinical Findings	1,241	General Health
Infectious	1,025	Immunity & Infections
Gastroenterology	970	Digestion & Gut
External Causes / Prevention	909	Safety & Prevention
Mental Health	858	Brain & Mind
Neurology	842	Brain & Mind
Dermatology	785	Skin & Hair

Clinical Domain	Disease Count	Frontend Tile
Ophthalmology	703	Eye Health
Endocrine/Metabolic	637	Metabolism & Hormones
Neonatology	625	Neonatal Health *(infants only)*
Cardiovascular	583	Heart & Circulation
Genitourinary	545	Kidneys & Urinary
Obstetrics	522	Women's Health *(females only)*
MSK/Rheumatology	425	Bones & Joints
Respiratory	344	Lungs & Breathing
Hematology	262	Blood & Immunity
Immunology	257	Immunity & Infections
ENT	151	Ear, Nose & Throat
Sleep Medicine	79	Brain & Mind
Sexual Health	68	Women's Health *(females only)*
TOTAL	**~16,383**	**23 domains → 19 tiles**

4 Step 1 — Building the Patient Vector

Translating every patient answer into a set of active database columns

When a patient fills out their personal details, the system converts every answer into a **set of active database columns** — called the **Patient Vector**.

📌 **Concrete example**

Patient says: "I am 35 years old, male, I smoke, and my father had heart disease."

System translates to: `{ age_26_to_39_years_risk, male, addiction_tobacco_smoking_rf, famhist_father_rf }`

Category 1 — Age

Patient's Age Group	DB Column Activated
0–28 days	`age_0_28_days_risk`
29 days to 1 year	`age_29_days_to_1_year_risk`
1 year to 3 years	`age_1_to_3_years_risk`
4 years to 5 years	`age_4_to_5_years_risk`
6 years to 12 years	`age_6_to_12_years_risk`
13 years to 17 years	`age_13_to_17_years_risk`
18 years to 25 years	`age_18_to_25_years_risk`
26 years to 39 years	`age_26_to_39_years_risk`
40 years to 59 years	`age_40_to_59_years_risk`
60 years to 74 years	`age_60_to_74_years_risk`
75 years and above (Geriatric)	`age_75_years_and_above_risk`

Category 2 — Vaccinations

✅ **Only "Missing" vaccinations add risk**

A completed vaccine is protective and contributes zero risk score. Only vaccination gaps (status = "Missing") are activated in the patient vector. 21 vaccines are tracked.

Vaccine	DB Column (activated only if Missing)
BCG	`bcg_missing_score`
OPV (Birth Dose)	`opv_0_missing_score`
Hepatitis B (Birth)	`hepb_birth_missing_score`
DPT	`dpt_missing_score`
IPV	`ipv_missing_score`
Hib	`hib_missing_score`
Rotavirus	`rotavirus_missing_score`
PCV	`pcv_missing_score`
MMR	`mmr_missing_score`
Varicella	`varicella_missing_score`
Hepatitis A	`hepa_missing_score`
DPT Booster	`dpt_booster_missing_score`
OPV Booster	`opv_booster_missing_score`
Tdap	`tdap_missing_score`
Influenza	`influenza_missing_score`
HPV	`hpv_missing_score`
COVID-19	`covid19_missing_score`
Td Booster	`td_booster_missing_score`
Zoster	`zoster_missing_score`
Pneumococcal	`pneumococcal_missing_score`
COVID-19 Booster	`covid19_booster_missing_score`

Category 3 — Gender

Maps to one of three columns: `male` , `female` , or `other` .

Category 4 — BMI (Auto-Calculated)

The patient provides height (cm) and weight (kg). BMI is always computed by the system — never self-reported.

$$BMI = weight_kg \div (height_m)^2$$

Underweight (<18.5) · Normal (18.5–24.9) · Overweight (25–29.9) · Obese Class I (30–34.9) · Obese Class II (35–39.9) · Obese Class III (≥40)

BMI Range	DB Column
< 18.5 (Underweight)	`bmi_underweight_rf`
18.5 – 24.9 (Normal)	`bmi_normal_weight_rf`
25 – 29.9 (Overweight)	`bmi_overweight_rf`
30 – 34.9 (Obese Class I)	`bmi_obesity_class_i_rf`
35 – 39.9 (Obese Class II)	`bmi_obesity_class_ii_rf`
≥ 40 (Obese Class III)	`bmi_obesity_class_iii_rf`

Categories 5–19 — Summary Table

#	Category	Options	DB Column Pattern
5	Blood Group	A+, A-, B+, B-, AB+, AB-, O+, O-, Unknown	`bg_[type]_rf`
6	Ethnicity	12 options incl. Asian, ANI, ASI, Caucasian, etc.	`eth_[ethnicity]_rf`
7	Residence History	16 countries	`residence_[country]_rf`

#	Category	Options	DB Column Pattern
8	Travel History	16 countries (same as residence)	`travel_[country]_rf`
9	Pregnancy / Fertility	Stage of life = Pregnancy or IVF/IUI/Ovulation active	`pregnancy_risk`
10	Occupational Exposures	13 types (dust, radiation, asbestos, shift work, etc.)	`occexp_[type]_rf`
11	Existing Conditions	12 comorbidities (hypertension, diabetes, heart disease, etc.)	`comorb_[condition]_rf`
12	Past Surgeries	14 types (orthopedic, cardiac, ENT, cataract, etc.)	`surg_[type]_rf`
13	Current Medications	19 categories (steroids, immunosuppressants, chemo, etc.)	`med_[type]_rf`
14	Known Allergies	11 types (penicillin, food, dust, latex, etc.)	`allergy_[type]_rf`
15	Addictions / Substance Use	11 types (smoking, alcohol, opioids, betel nut, etc.)	`addiction_[type]_rf`
16	Injury History	10 types (fracture, head injury, burn, poisoning, etc.)	`injury_[type]_rf`
17	Co-living Situations	14 types (hostel, prison, near industrial zone, etc.)	`coliving_[type]_rf`
18	Diet Patterns	10 types (red meat, processed foods, high salt, etc.)	`diet_[type]_rf`

#	Category	Options	DB Column Pattern
19	Pets at Home	10 types (dog, cat, bird, cow/buffalo, etc.)	`pet_[type]_diseasespecific_prob_0_`
20	Family Health History	17 conditions × 7 relation groups	Handled separately — see Section 7

5 Step 2 — Domain Scoring

Running the patient vector against 16,000+ diseases in a single SQL query

Once the patient vector is built, a single SQL query runs against the entire `disease_risk_scores` table:

```
SELECT
  clinical_domain,
  AVG(col_1 + col_2 + col_3 + ... all active cols except famhist_*) AS avg_score,
  AVG(age_col + gender_col)  AS avg_baseline
FROM disease_risk_scores
WHERE clinical_domain IS NOT NULL AND clinical_domain ≠ ''
GROUP BY clinical_domain
ORDER BY avg_score DESC
```

What this computes: For each clinical domain, take every disease in that domain, sum all the active risk factor column values for that disease, and take the average across all diseases in the domain.

🔬 **Concrete example — Endocrine/Metabolic domain**

Patient: Age 35, male, Diabetes as comorbidity, high-sugar diet. For each of the 637 Endocrine diseases, the system sums:

`age_26_to_39_years_risk + male + comorb_diabetes_mellitus_rf + diet_sugary_foods_beverages_rf`

The average of those 637 sums = **avg_score** for Endocrine/Metabolic.

The domain with the highest `avg_score` is the one where this patient's specific combination of factors most consistently elevates risk across the widest range of diseases in that domain.

6 Step 3 — Excess Scoring (The Fairness Layer)

The most important architectural decision — making domain ranking fair

⚠️ **The problem with raw scores**

A 65-year-old will score high in Cardiovascular simply because the age column has high values across 583 cardiovascular diseases. MSK has only 425. So Cardiovascular *always* dominates in raw scoring — even if the patient has zero cardiovascular-specific risk factors. This is a mathematical artifact, not a real clinical signal.

The Excess Score Formula

```
excess_score = (patient_score - baseline_score) + family_boost
```

patient_score = AVG score with all active columns (excluding famhist_*)
baseline_score = AVG score with ONLY age + gender columns
family_boost = condition-specific family history boost (Section 7)

What baseline_score represents: The score that *any generic person of this age and gender* would receive in this domain — no lifestyle factors, no conditions, no diet. Just age and gender.

What excess_score represents: How much THIS patient scores above and beyond a generic person of the same age and gender. It answers: "Is this person's risk elevated because of who they are and how they live, or just because of their age?"

Side-by-Side Example

Domain	Patient Score (raw)	Baseline Score	Excess Score	Result
Cardiovascular	14.2	13.8	**0.4**	Deprioritized
MSK/Rheumatology	3.1	0.5	**2.6**	Flagged

The patient smokes and has had orthopedic surgery — factors that are specifically high in MSK diseases. Even though Cardiovascular has the higher *raw* score, MSK/Rheumatology is correctly identified as the area of genuine patient-specific elevated risk.

✅ **All domains are sorted and ranked by excess_score, not raw score.**

This ensures the ranking always reflects the patient's actual specific risk profile, never the mathematical structure of the database.

7 Step 4 — Family History Special Treatment

Why family history needs its own engine — and how condition-specific routing works

The Problem with Generic Family History Columns

The database contains 11 generic `famhist_*` columns (father, mother, siblings, paternal grandfather, paternal grandmother, maternal grandfather, maternal grandmother, paternal uncles, paternal aunts, maternal uncles, maternal aunts).

These encode **generic heritability** — the statistical fact that having any relative with any disease slightly increases risk for every disease. The problem: Cardiovascular has the highest heritability coefficients. So if a patient's father has *Bone Disease*, the raw system would boost Cardiovascular the most — which is clinically wrong and dangerous.

The Solution — Condition-Specific Domain Routing

`famhist_*` columns are **completely excluded from domain scoring**. Instead, each specific family condition is routed to boost the correct clinical domain:

Family Condition Selected	Domain(s) Boosted
Heart Disease / Cardiovascular	Cardiovascular
Hypertension	Cardiovascular
Stroke	Cardiovascular, Neurology
Diabetes	Endocrine/Metabolic
Thyroid Disorder	Endocrine/Metabolic
Cancer / Neoplasms	Oncology
Mental Health Disorders	Mental Health
Kidney Disease	Genitourinary
Liver Disease	Gastroenterology
Digestive Disorders	Gastroenterology
Asthma / Respiratory Disease	Respiratory
Neurological Disorders	Neurology
Bone / Joint Disorders	MSK/Rheumatology

Family Condition Selected	Domain(s) Boosted
Eye Disorders	Ophthalmology
Blood Disorders	Hematology
Skin Disorders	Dermatology
Infectious Diseases	Infectious

The Boost Formula

```
boost = (affected_relations ÷ 7) × MAX_FAMILY_BOOST (8)
```

7 = total possible relation groups · MAX_FAMILY_BOOST = 8.0 · Multiple conditions stack independently per domain

Relations Affected	Boost Amount	Clinical Interpretation
1 of 7	1.14	One first-degree or second-degree relative
2 of 7	2.29	Two relation groups (e.g., father + siblings)
3 of 7	3.43	Strong family pattern emerging
4 of 7	4.57	Clear familial predisposition
5 of 7	5.71	Strongly hereditary pattern
6 of 7	6.86	Near-universal family history
7 of 7 (all)	8.00	Maximum — all relation groups affected

🧬 **Why MAX_FAMILY_BOOST = 8?**

The typical excess score spread across domains for an average patient peaks at ~2.2. A MAX_FAMILY_BOOST of 8 ensures that even moderate family history (3–4 affected relation groups, boost ~3.4–4.6) clearly overcomes normal demographic variation and correctly flags the domain. A previous lower value (3.5) was insufficient — it could be drowned out by other factors, creating false negatives.

8 Step 5 — The "Worth Monitoring" Threshold

A self-calibrating statistical threshold that adapts to each patient's own score distribution

```
threshold = mean(all excess_scores) + 0.5 × SD(all excess_scores)
```

Any domain with excess_score ≥ threshold is flagged "Worth monitoring"

Why Not a Fixed Threshold?

Approach	Problem	Chosen?
Fixed cutoff (e.g., score > 2.0)	Breaks for patients with few details (all scores low, nothing flagged) OR many details (all scores high, everything flagged)	❌ No
Always show top N domains	Flags N domains even for genuinely healthy patients — creates baseless fear	❌ No
Mean + 0.5 SD (self-calibrating)	Adapts to the actual spread of each patient's scores. Conservative (0.5 SD) honors "false negative is dangerous" principle.	✅ Yes

Full Numerical Example

📊 Threshold Calculation

Patient's excess scores across all domains:

```
Cardiovascular:      0.4
MSK/Rheumatology:    2.6
Endocrine/Metabolic: 1.8
Respiratory:         0.3
```

```
Mental Health:       0.2
Oncology:            1.1
```

- **Mean** = (0.4 + 2.6 + 1.8 + 0.3 + 0.2 + 1.1) / 6 = **1.07**
- **SD** ≈ **0.83**
- **Threshold** = 1.07 + (0.5 × 0.83) = **1.48**

Domains ≥ 1.48: **MSK/Rheumatology (2.6)** and **Endocrine/Metabolic (1.8)**

Patient sees "Worth monitoring" on **Bones & Joints** and **Metabolism & Hormones**. All others show "Looking good".

9 Step 6 — Frontend Display & Age/Gender Gates

How scores become the visual risk profile the patient sees

Age Gate — Neonatal Health Tile

The **Neonatal Health** tile is only shown if the patient's age is **0–28 days** or **29 days to 1 year**. For any patient 1 year or older, this tile is hidden entirely. Reason: Neonatology diseases have non-zero values for adult age columns (DB artifact), which would cause incorrect flagging for adult patients.

Gender Gate — Women's Health Tile

The **Women's Health** tile (Obstetrics + Sexual Health) is hidden for male patients. These domains are biologically specific to female anatomy. Showing it to males would erode trust and create confusion.

How Many Tiles a Patient Sees

Patient Type	Tiles Visible
Adult Male (≥ 1 year)	17 (19 − Neonatal − Women's Health)
Adult Female (≥ 1 year)	18 (19 − Neonatal)
Male Infant (0–1 year)	18 (19 − Women's Health)
Female Infant (0–1 year)	19 (all tiles)

Tile Multi-Domain Logic

Some tiles cover multiple DB domains. If **any** domain within a tile's list crosses the threshold, the tile shows "Worth monitoring".

Tile Name	Domains Covered
Brain & Mind	Neurology, Mental Health, Sleep Medicine
Immunity & Infections	Infectious, Immunology
Women's Health	Obstetrics, Sexual Health
All other tiles	*One domain each*

Paid vs Free Users

Feature	Free User	Premium User
See "Worth monitoring" / "Looking good" badge	✅ Yes	✅ Yes
Top 3 reasons in plain English	❌ Locked	✅ Yes
Category breakdown bar chart	❌ Locked	✅ Yes
Upgrade prompt on flagged tiles	✅ Shown	N/A

10 Step 7 — Stale Detection & Auto-Recomputation

Ensuring patients always see current, correct data

Every stored risk profile in `patient_risk_profile` contains a `domain_scores` JSONB field. When a patient loads their profile, the system checks:

```
const isStale = !existing
  || existing.domain_scores === null
  || (Array.isArray(existing.domain_scores)
      && existing.domain_scores.length > 0
      && existing.domain_scores[0].excess_score === undefined);
```

If the profile is stale — either missing entirely, or computed before excess scoring was introduced (identifiable by the absence of the `excess_score` field) — the system **automatically recomputes** the profile on the server before returning it to the patient.

🔄 **Recalculate Button**

Every patient also has a **Recalculate** button in the UI that forces an immediate full recomputation. Patients should use this after updating any section of their personal details, or after filling in details for the first time.

11 Top Diseases (ICD-11 Level Detail)

Individual disease scoring stored alongside domain scores

In addition to the domain-level view, the engine computes and stores the **top 20 highest-scoring individual diseases** for each patient (by raw patient score). Each disease entry includes:

```
{
  "icd11":            "BA00",
  "title":            "Type 2 Diabetes Mellitus",
  "clinical_domain":  "Endocrine/Metabolic",
  "chapter":          "Endocrine, nutritional or metabolic diseases",
  "score":            3.42,
  "category_breakdown": {
    "Age": 0.8, "BMI": 1.2, "Diet": 0.9, "Family": 0.52
  }
}
```

This data is stored in `patient_risk_profile.top_diseases` and is used by the **symptom checker** feature for cross-referencing. Individual disease names are intentionally not shown on the main risk profile screen — showing "Type 2 Diabetes" as a top-scored disease to a patient who is not diabetic can cause unnecessary anxiety. The domain-level view (e.g., "Metabolism & Hormones: Worth monitoring") is the appropriate level of granularity for lay users.

12 All 23 Clinical Domains & 19 Body System Tiles

Complete mapping — every DB domain covered, zero silent misses

#	DB Clinical Domain (exact string)	Disease Count	Frontend Tile	Tile Key
1	Injury/Toxicology	1,982	Injury & Toxins	`injury`
2	Congenital/Genetics	1,323	Genetics & Inherited	`genetics`
3	Oncology	1,247	Cell Health	`cell`
4	Undifferentiated Clinical Findings	1,241	General Health	`general`
5	Infectious	1,025	Immunity & Infections	`immunity`
6	Gastroenterology	970	Digestion & Gut	`digestion`
7	External Causes / Prevention	909	Safety & Prevention	`prevention`
8	Mental Health	858	Brain & Mind	`brain`
9	Neurology	842	Brain & Mind	`brain`
10	Dermatology	785	Skin & Hair	`skin`
11	Ophthalmology	703	Eye Health	`eyes`
12	Endocrine/Metabolic	637	Metabolism & Hormones	`metabolism`
13	Neonatology	625	Neonatal Health *(infants only)*	`neonatal`
14	Cardiovascular	583	Heart & Circulation	`heart`
15	Genitourinary	545	Kidneys & Urinary	`urinary`
16	Obstetrics	522	Women's Health *(females only)*	`womens`

#	DB Clinical Domain (exact string)	Disease Count	Frontend Tile	Tile Key
17	MSK/Rheumatology	425	Bones & Joints	`bones`
18	Respiratory	344	Lungs & Breathing	`lungs`
19	Hematology	262	Blood & Immunity	`blood`
20	Immunology	257	Immunity & Infections	`immunity`
21	ENT	151	Ear, Nose & Throat	`ent`
22	Sleep Medicine	79	Brain & Mind	`brain`
23	Sexual Health	68	Women's Health *(females only)*	`womens`

✅ **All 23 DB domains are covered. Zero are silently dropped.**

This was verified by running `SELECT DISTINCT clinical_domain FROM disease_risk_scores ORDER BY clinical_domain` and cross-checking every result against the frontend tile list. The one mismatch found and fixed: the DB uses `"External Causes / Prevention"` (with spaces around the slash), not `"External Causes/Prevention"`.

13 Complete Worked Example

Step-by-step walkthrough with a real patient scenario

👤 Patient: Ravi, Adult Male

AGE GROUP 40–59 years	**GENDER** Male
BMI 28.5 — Overweight	**ADDICTIONS** Tobacco/Smoking, Alcohol
DIET Red meat, High salt	**COMORBIDITIES** Hypertension
MEDICATIONS Antihypertensives	**MISSING VACCINES** COVID-19, Influenza
OCCUPATION Shift Work / Night Duty	**FAMILY HISTORY** Heart Disease: Father, Mother · Diabetes: Father

Step 1 — Patient Vector

```
Active columns:
age_40_to_59_years_risk        ← age
male                           ← gender
bmi_overweight_rf              ← BMI 28.5 (auto-calculated)
addiction_tobacco_smoking_rf   ← smoking
addiction_alcohol_use_rf       ← alcohol
diet_red_meat_rf               ← diet
diet_salt_processed_sodium_rf  ← diet
comorb_hypertension_rf         ← existing condition
med_antihypertensives_rf       ← medication
covid19_missing_score          ← missing vaccine
influenza_missing_score        ← missing vaccine
occexp_shiftwork_nightduty_rf  ← occupation
famhist_father_rf              ← family (EXCLUDED from domain SQL)
famhist_mother_rf              ← family (EXCLUDED from domain SQL)
```

Step 2 — Family Boosts

```
Heart Disease / Cardiovascular (Father + Mother = 2 of 7 relations):
  boost = (2 ÷ 7) × 8 = 2.29  → added to Cardiovascular

Diabetes (Father only = 1 of 7 relations):
  boost = (1 ÷ 7) × 8 = 1.14  → added to Endocrine/Metabolic
```

Step 3 — Domain Scores

Domain	Patient Score	Baseline	Excess Before Family	Family Boost	Final Excess Score
Cardiovascular	**18.3**	**14.1**	**4.2**	**+2.29**	**6.49**
Endocrine/Metabolic	**5.1**	**2.8**	**2.3**	**+1.14**	**3.44**
Respiratory	4.2	1.6	2.6	0	2.60
Gastroenterology	3.8	1.8	2.0	0	2.00
MSK/Rheumatology	2.1	0.8	1.3	0	1.30
Mental Health	1.8	0.9	0.9	0	0.90
Oncology	1.4	0.7	0.7	0	0.70
Others...	—	—	—	0	≤ 0.50

Highlighted rows = flagged as "Worth monitoring"

Step 4 — Threshold

```
Mean of all excess scores = 1.82
SD = 1.76
Threshold = 1.82 + (0.5 × 1.76) = 2.70

Domains ≥ 2.70:
  ✅ Cardiovascular     6.49  → Heart & Circulation: Worth Monitoring
  ✅ Endocrine/Metabolic 3.44  → Metabolism & Hormones: Worth Monitoring
```

Step 5 — What Ravi Sees

❤️ **Heart & Circulation**
Worth monitoring

⚡ **Metabolism & Hormones**
Worth monitoring

🫁 **Lungs & Breathing**
Looking good

🧠 **Brain & Mind**
Looking good

🦴 **Bones & Joints**
Looking good

🛡️ **Immunity & Infections**
Looking good

+ 11 more tiles
Looking good

💬 **What Ravi would read (paid tier)**

"Heart & Circulation showed up for you because of your lifestyle habits, existing health conditions, and family history."

14 All Input Categories & DB Column Mappings

Category subtotal columns used for breakdown charts

In addition to the individual factor columns, the DB has precomputed category subtotal columns for each disease. These are used to construct the percentage breakdown chart shown to premium users inside each flagged tile — telling them which category of their health profile contributed most to that domain's risk.

Category	DB Subtotal Column	What It Represents
Age	`sub_age_risk`	Sum of all age-related risk for this disease
Vaccine	`sub_vaccine_missing`	Sum of all missing-vaccine risk
Gender	`sub_gender`	Gender-specific risk contribution
Blood Group	`sub_blood_group_risk`	Blood-type-specific risk
Ethnicity	`sub_ethinicity_risk`	Ethnicity-specific risk
Residence	`sub_residence_risk`	Geographic residence risk
Travel	`sub_travel_risk`	Travel history risk
BMI	`sub_bmi_risk`	Weight-status risk
Occupation	`sub_occupation_risk`	Occupational exposure risk
Comorbidity	`sub_comorbidity_risk`	Existing condition risk
Surgery	`sub_surgery_risk`	Surgical history risk
Medication	`sub_medication_risk`	Current medication risk
Allergy	`sub_allergy_risk`	Allergy-related risk
Addiction	`sub_addiction_risk`	Substance use risk
Injury	`sub_injury_risk`	Injury history risk
Coliving	`sub_coliving_risk`	Co-living environment risk
Diet	`sub_diet_risk`	Dietary pattern risk

Category	DB Subtotal Column	What It Represents
Pet	`sub_pet_risk`	Pet exposure risk
Family	*Computed via family boost — no subtotal column*	Family condition-specific boost

15 Key Design Decisions & Why

The reasoning behind every major architectural choice

1️⃣ **Excess score over raw score**

Raw scores always favor large domains. A 65-year-old will almost always have high cardiovascular raw scores regardless of their lifestyle. Excess scoring isolates what is truly patient-specific. This is the single most important architectural decision in the engine.

2️⃣ **Family history completely stripped from domain scoring**

The `famhist_*` columns in the DB encode generic heritability that correlates most with cardiovascular diseases. Using them for domain scoring would give wrong boosts to wrong domains. The condition-specific routing map gives precise, clinically meaningful boosts to the right body systems.

3️⃣ **MAX_FAMILY_BOOST = 8**

The typical excess score spread across domains for an average patient peaks at ~2.2. MAX_FAMILY_BOOST = 8 means even moderate family history (3–4 relations, boost ~3.4–4.6) clearly overcomes baseline demographic noise. A smaller value (3.5, the previous setting) was drowned out by other factors, creating false negatives.

4️⃣ **Threshold = Mean + 0.5 SD (self-calibrating)**

A fixed threshold would fail for sparse and dense profiles alike. The statistical approach self-calibrates per patient. The 0.5 SD factor (below the more common 1 SD cutoff) keeps the system conservative and aligned with "false negative is dangerous."

5️⃣ **All 23 DB domains must have frontend tiles**

Any unmapped domain is a silent false negative. With 909 diseases in "External Causes / Prevention" — a real domain — failing to show it would mean a patient with toxic occupational exposure never gets warned. We verified all 23 domains by running a DB query and matching every result to the tile list.

6️⃣ **Age gate (Neonatal) and gender gate (Women's Health)**

Neonatology has non-zero values for adult age columns — a DB artifact that would incorrectly flag the tile for adults. The age gate prevents this. Women's Health covers obstetric/reproductive conditions irrelevant to males; showing it would erode patient trust.

7️⃣ **Vaccines: only "Missing" status adds risk**

A completed vaccine is protective and contributes zero risk. Only vaccination gaps represent a risk factor. Treating completed vaccines as risk-increasing would be clinically incorrect and counter-intuitive.

8️⃣ **BMI is always calculated, never self-reported**

Self-reported weight is notoriously inaccurate. The system always computes BMI from height and weight and maps it to the correct DB column internally. The patient never selects their own BMI category.

16

What the Risk Profile Is NOT

Critical boundaries every stakeholder must understand

🚫 **It is not a diagnosis**

The risk profile identifies elevated statistical risk, not the presence of a disease. A patient flagged for "Heart & Circulation" does not have heart disease — they have a combination of factors statistically associated with cardiovascular diseases.

🚫 **It is not a medical recommendation**

The system explicitly shows on-screen: "This is not a diagnosis." The purpose is to encourage patients to discuss flagged areas with their doctor, not to self-treat or self-diagnose.

🚫 **It is not real-time clinical data**

The system uses self-reported information from the patient's personal details form. It does not use lab results, imaging, vitals, or clinical examination data. Accuracy depends entirely on how completely and accurately the patient fills in their details.

🚫 **It is not a substitute for clinical judgment**

A doctor reviewing the patient should use this as a starting point for conversation, not a clinical decision tool. Two patients with similar profiles may get slightly different results based on the spread of their scores.

17

Data Flow — End-to-End Summary

From form fill to the patient's screen, every step

1

Patient Fills Personal Details

19 health categories collected: age, gender, BMI, vaccines, diet, medications, family history, and 12 more.

2

buildPatientVector()

Every answer is translated to a Set of active DB column names (~5–40 columns depending on how much the patient filled in). Family history columns are included in the Set but flagged for separate handling.

3

SQL Query → disease_risk_scores

For each clinical domain: AVG(active cols except famhist_*) = avg_score · AVG(age + gender cols) = avg_baseline. Groups 16,000+ diseases into 23 domain averages in a single query.

4

Family History Boost Calculation

Each family condition → FAMILY_CONDITION_DOMAIN_MAP → boost = (relations/7) × 8. Boosts applied per clinical domain directly.

5

Excess Score Computation

excess_score = (avg_score − avg_baseline) + family_boost. All 23 domains sorted by excess_score descending.

6

Stored in patient_risk_profile (PostgreSQL)

UPSERT into patient_risk_profile table. Stores: domain_scores (JSONB with excess_score + category breakdown per domain), top_diseases (top 20 ICD-11 diseases by raw score), total_diseases_scored, computed_at timestamp.

7

Frontend Loads Profile

Stale check runs. If stale, recompute triggered automatically. Profile returned to browser.

8

Threshold Calculated Client-Side

threshold = mean(excess_scores) + 0.5 × SD(excess_scores). worthMonitoring = Set of domains ≥ threshold.

9

17–19 Tiles Rendered (gates applied)

Each tile checks if any of its covered domains is in worthMonitoring. Age gate hides Neonatal for patients ≥ 1 year. Gender gate hides Women's Health for males.

✓

Patient Sees Their Personalized Risk Map

Each tile: "Worth monitoring" (amber badge) or "Looking good" (green badge). Premium users get drill-down: top 3 reasons + category breakdown chart.

Cura Health — Risk Profile Engine | Version: Launch-Locked v1.0 | April 2026

Risk profile logic frozen at this version for launch and post-launch operations.
Any future changes to scoring logic require full recomputation of all stored patient profiles.

This document is for internal and stakeholder reference only. The risk profile is a statistical screening tool, not a clinical diagnosis. Always consult a qualified physician.